Héber H. Arcolezi, Ph.D.

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Professional Experience

Feb 2026 – Present	**Tenure-Track Assistant Professor**, ÉTS Montréal, Canada.
Oct 2023 – Jan 2026	**Tenured Research Scientist (ISFP)**[1], Inria Grenoble, France.
Feb 2022 – Sept 2023	**Postdoctoral Researcher**, Inria Saclay & École Polytechnique, France.

Education

Jun 2019 – Jan 2022	**Ph.D. in Computer Science**, Université Bourgogne Franche-Comté (UBFC) **Thesis:** *Production of Categorical Data Verifying Differential Privacy: Conception and Applications to Machine Learning* [link] **Advisors:** Jean-François Couchot, Bechara Al Bouna, and Xiaokui Xiao
Sep 2017 – Aug 2019	**M.Sc. in Electrical Engineering**, São Paulo State University (UNESP) **Thesis:** *A Novel Robust and Intelligent Control Based Approach for Human Lower Limb Rehabilitation via Neuromuscular Electrical Stimulation* [link] **Advisor:** Aparecido A. de Carvalho
Jul 2012 – Aug 2017	**B.Eng. in Electrical Engineering**, Mato Grosso State University (UNEMAT) **Thesis:** *Um Estudo Complementar ao Projeto de Controle PID no Caso do Pêndulo Invertido (in Portuguese)* [link] **Advisor:** Rogério B. Quirino

Publications

Conference Proceedings

1. **H. H. Arcolezi**, "Private frequency estimation via residue number systems," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 35 518–35 526. DOI: 10.1609/aaai.v40i42.40862.
2. **H. H. Arcolezi** and S. Gambs, "Revisiting locally differentially private protocols: Towards better trade-offs in privacy, utility, and attack resistance," in *2026 IEEE 42nd International Conference on Data Engineering (ICDE)*, 2026, pp. 294–307. DOI: 10.1109/ICDE65706.2026.00029.
3. M. Babaei, Y. Wang, H. Lautraite, **H. H. Arcolezi**, U. Aïvodji, and S. Gambs, "Quantifying the privacy of counterfactuals by leveraging membership inference attacks against synthetic data," in *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT '26, New York, NY, USA: Association for Computing Machinery, 2026, pp. 4988–5022, ISBN: 9798400725968. DOI: 10.1145/3805689.3812361.
4. R. G. Gonze, N. Fernandes, **H. H. Arcolezi**, C. Palamidessi, and N. Bielova, " Beyond Epsilon: A Principled QIF Framework for Local Differential Privacy," in *2026 39th IEEE Computer Security Foundations Symposium (CSF) CSF*, Los Alamitos, CA, USA: IEEE Computer Society, Jul. 2026, pp. 483–496. DOI: 10.1109/CSF68417.2026.00032.
5. A. A. Marreiras Neto, **H. H. Arcolezi**, and J. C. Machado, "Data poisoning in longitudinal local differential privacy: Attacks and analysis," in *Data and Applications Security and Privacy XL*, Springer Nature Switzerland, 2026, pp. 124–141, ISBN: 9783032332608. DOI: 10.1007/978-3-032-33260-8_7.

[1]Inria ISFP researchers have a light teaching load, ranging from approximately 32 to 64 hours per year. More information at https://www.inria.fr/fr/inria-starting-faculty-position-isfp.

6 A. K. Mishra, M. Cunche, and **H. H. Arcolezi**, "How tough is location anonymization? re-identifying 100k real-user trajectories in japan," in *Proceedings of the ACM Asia Conference on Computer and Communications Security*, ser. ASIA CCS '26, Bangalore, India: Association for Computing Machinery, 2026, pp. 1261–1273, ISBN: 9798400723568. DOI: 10.1145/3779208.3805978.

7 C. A. Pinzón, E. ElSalamouny, L. Massot, A. Miller, **H. H. Arcolezi**, and C. Palamidessi, "Estimating the true distribution of data collected with randomized response," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 35 751–35 758. DOI: 10.1609/aaai.v40i42.40888.

8 **H. H. Arcolezi**, M. Alishahi, A.-A. Bendoukha, and N. Kaaniche, "Fair play for individuals, foul play for groups? auditing anonymization's impact on ml fairness," in *ECAI 2025*, IOS Press, Oct. 2025. DOI: 10.3233/faia250909.

9 A.-A. Bendoukha, **H. H. Arcolezi**, N. Kaaniche, A. Boudguiga, R. Sirdey, and P.-E. Clet, "Fade: Federated aggregation with discrimination elimination," in *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT '25, Association for Computing Machinery, 2025, pp. 3182–3195, ISBN: 9798400714825. DOI: 10.1145/3715275.3732203.

10 C. Eichler, N. Champeil, N. Anciaux, A. Bensamoun, **H. H. Arcolezi**, and J. M. De Fuentes, "Nob-MIAs: Non-biased membership inference attacks assessment on large language models with ex-post dataset construction," in *Web Information Systems Engineering – WISE 2024*, Singapore: Springer Nature Singapore, 2025, pp. 441–456. DOI: 10.1007/978-981-96-0570-5_32.

11 H. Zhang, A. K. Mishra, and **H. H. Arcolezi**, "Demo: Exploring utility and attackability trade-offs in local differential privacy," in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '25, Taipei, Taiwan: Association for Computing Machinery, 2025, pp. 4728–4730. DOI: 10.1145/3719027.3760706.

12 R. Binkyte, C. A. Pinzón, S. Lestyán, K. Jung, **H. H. Arcolezi**, and C. Palamidessi, "Causal discovery under local privacy," in *Proceedings of the Third Conference on Causal Learning and Reasoning*, F. Locatello and V. Didelez, Eds., ser. Proceedings of Machine Learning Research, vol. 236, PMLR, Jan. 2024, pp. 325–383. URL: https://proceedings.mlr.press/v236/binkyte24a.html.

13 K. Makhlouf, T. Stefanović, **H. H. Arcolezi**, and C. Palamidessi, "A systematic and formal study of the impact of local differential privacy on fairness: Preliminary results," in *2024 IEEE 37th Computer Security Foundations Symposium (CSF)*, 2024, pp. 1–16. DOI: 10.1109/CSF61375.2024.00039.

14 **H. H. Arcolezi**, S. Cerna, and C. Palamidessi, "On the utility gain of iterative bayesian update for locally differentially private mechanisms," in *Data and Applications Security and Privacy XXXVII*, Springer Nature Switzerland, 2023, pp. 165–183, ISBN: 9783031375866. DOI: 10.1007/978-3-031-37586-6_11.

15 **H. H. Arcolezi**, K. Makhlouf, and C. Palamidessi, "(local) differential privacy has no disparate impact on fairness," in *Data and Applications Security and Privacy XXXVII*, V. Atluri and A. L. Ferrara, Eds., Cham: Springer Nature Switzerland, 2023, pp. 3–21. DOI: 10.1007/978-3-031-37586-6_1.

16 **H. H. Arcolezi**, C. A. Pinzón, C. Palamidessi, and S. Gambs, "Frequency estimation of evolving data under local differential privacy," in *Proceedings of the 26th International Conference on Extending Database Technology, EDBT 2023, Ioannina, Greece, March 28 - March 31, 2023*, OpenProceedings.org, 2023, pp. 512–525. DOI: 10.48786/EDBT.2023.44.

17 **H. H. Arcolezi**, J.-F. Couchot, S. Gambs, C. Palamidessi, and M. Zolfaghari, "Multi-freq-ldpy: Multiple frequency estimation under local differential privacy in python," in *Computer Security – ESORICS 2022*, V. Atluri, R. Di Pietro, C. D. Jensen, and W. Meng, Eds., Cham: Springer Nature Switzerland, 2022, pp. 770–775. DOI: 10.1007/978-3-031-17143-7_40.

18 **H. H. Arcolezi**, J.-F. Couchot, B. Al Bouna, and X. Xiao, "Random sampling plus fake data: Multidimensional frequency estimates with local differential privacy," in *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*, ACM, Oct. 2021, pp. 47–57. DOI: 10.1145/3459637.3482467.

19 **H. H. Arcolezi**, J.-F. Couchot, B. A. Bouna, and X. Xiao, "Longitudinal collection and analysis of mobile phone data with local differential privacy," in *IFIP International Summer School on Privacy and Identity Management*, Springer International Publishing, 2021, pp. 40–57, ISBN: 9783030724658. DOI: 10.1007/978-3-030-72465-8_3.

20 **H. H. Arcolezi**, J.-F. Couchot, O. Baala, J.-M. Contet, B. Al Bouna, and X. Xiao, "Mobility modeling through mobile data: Generating an optimized and open dataset respecting privacy," in *2020 International Wireless Communications and Mobile Computing (IWCMC)*, 2020, pp. 1689–1694. DOI: 10.1109/IWCMC48107.2020.9148138.

21 S. Cerna, C. Guyeux, **H. H. Arcolezi**, R. Couturier, and G. Royer, "A comparison of lstm and xgboost for predicting firemen interventions," in *Trends and Innovations in Information Systems and Technologies*, Springer International Publishing, 2020, pp. 424–434, ISBN: 9783030456917. DOI: 10.1007/978-3-030-45691-7_39.

22 S. Cerna, C. Guyeux, **H. H. Arcolezi**, and G. Royer, "Boosting methods for predicting firemen interventions," in *2020 11th International Conference on Information and Communication Systems (ICICS)*, 2020, pp. 001–006. DOI: 10.1109/ICICS49469.2020.239488.

23 **H. H. Arcolezi**, W. R. B. M. Nunes, S. L. C. Ñahuis, M. A. A. Sanches, M. C. M. Teixeira, and A. A. de Carvalho, "A rise-based controller fine-tuned by an improved genetic algorithm for human lower limb rehabilitation via neuromuscular electrical stimulation," in *2019 6th International Conference on Control, Decision and Information Technologies (CoDIT)*, 2019, pp. 1197–1202. DOI: 10.1109/CoDIT.2019.8820357.

24 S. L. C. Ñahuis, C. Guyeux, **H. H. Arcolezi**, R. Couturier, G. Royer, and A. D. P. Lotufo, "Long short-term memory for predicting firemen interventions," in *2019 6th International Conference on Control, Decision and Information Technologies (CoDIT)*, 2019, pp. 1132–1137. DOI: 10.1109/CoDIT.2019.8820671.

Journal Articles

25 V. G. Angelozzi and **H. H. Arcolezi**, "Where to intervene? benchmarking fairness-aware learning on differentially private synthetic tabular data," *Proceedings on Privacy Enhancing Technologies*, vol. 2026, no. 4, pp. 53–82, Oct. 2026, ISSN: 2299-0984. DOI: 10.56553/popets-2026-0110.

26 P. Guerra-Balboa, A. Sauer, **H. H. Arcolezi**, and T. Strufe, "Understanding disclosure risk in differential privacy with applications to noise calibration and auditing," *Proceedings of the VLDB Endowment*, vol. 19, no. 7, pp. 1558–1571, Mar. 2026, ISSN: 2150-8097. DOI: 10.14778/3801059.3801069.

27 **H. H. Arcolezi**, K. Makhlouf, and C. Palamidessi, "Group fairness under obfuscated sensitive information," *Journal of Computer Security*, vol. 33, no. 4, pp. 239–260, May 2025, ISSN: 1875-8924. DOI: 10.1177/0926227x251330212.

28 **H. H. Arcolezi** and S. Gambs, "Revealing the true cost of locally differentially private protocols: An auditing perspective," *Proceedings on Privacy Enhancing Technologies*, vol. 2024, no. 4, pp. 123–141, 2024. DOI: 10.56553/popets-2024-0110.

29 K. Makhlouf, **H. H. Arcolezi**, S. Zhioua, G. B. Brahim, and C. Palamidessi, "On the impact of multi-dimensional local differential privacy on fairness," *Data Mining and Knowledge Discovery*, May 2024, ISSN: 1573-756X. DOI: 10.1007/s10618-024-01031-0.

30 **H. H. Arcolezi**, S. Gambs, J.-F. Couchot, and C. Palamidessi, "On the risks of collecting multidimensional data under local differential privacy," *Proc. VLDB Endow.*, vol. 16, no. 5, pp. 1126–1139, Jan. 2023, ISSN: 2150-8097. DOI: 10.14778/3579075.3579086.

31 **H. H. Arcolezi**, S. Cerna, J.-F. Couchot, C. Guyeux, and A. Makhoul, "Privacy-preserving prediction of victim's mortality and their need for transportation to health facilities," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 8, pp. 5592–5599, 2022. DOI: 10.1109/TII.2021.3123588.

32 **H. H. Arcolezi**, J.-F. Couchot, B. A. Bouna, and X. Xiao, "Improving the utility of locally differentially private protocols for longitudinal and multidimensional frequency estimates," *Digital Communications and Networks*, Jul. 2022. DOI: 10.1016/j.dcan.2022.07.003.

33 **H. H. Arcolezi**, J.-F. Couchot, D. Renaud, B. Al Bouna, and X. Xiao, "Differentially private multivariate time series forecasting of aggregated human mobility with deep learning: Input or gradient perturbation?" *Neural Computing and Applications*, vol. 34, no. 16, pp. 13 355–13 369, Jun. 2022, ISSN: 1433-3058. DOI: 10.1007/s00521-022-07393-0.

34 **H. H. Arcolezi**, S. Cerna, C. Guyeux, and J.-F. Couchot, "Preserving geo-indistinguishability of the emergency scene to predict ambulance response time," *Mathematical and Computational Applications*, vol. 26, no. 3, p. 56, Aug. 2021, ISSN: 2297-8747. DOI: 10.3390/mca26030056.

35 **H. H. Arcolezi**, W. R. Nunes, R. A. de Araujo, *et al.*, "Rise controller tuning and system identification through machine learning for human lower limb rehabilitation via neuromuscular electrical stimulation," *Engineering Applications of Artificial Intelligence*, vol. 102, p. 104 294, Jun. 2021, ISSN: 0952-1976. DOI: 10.1016/j.engappai.2021.104294.

36 S. Cerna, **H. H. Arcolezi**, C. Guyeux, G. Royer-Fey, and C. Chevallier, "Machine learning-based forecasting of firemen ambulances' turnaround time in hospitals, considering the covid-19 impact," *Applied Soft Computing*, vol. 109, p. 107 561, Sep. 2021, ISSN: 1568-4946. DOI: 10.1016/j.asoc.2021.107561.

37 **H. H. Arcolezi**, J.-F. Couchot, S. Cerna, *et al.*, "Forecasting the number of firefighter interventions per region with local-differential-privacy-based data," *Computers & Security*, vol. 96, p. 101 888, Sep. 2020, ISSN: 0167-4048. DOI: 10.1016/j.cose.2020.101888.

38 **H. H. Arcolezi**, W. R. B. M. Nunes, S. Cerna, *et al.*, "Identifying the knee joint angular position under neuromuscular electrical stimulation via long short-term memory neural networks," *Research on Biomedical Engineering*, vol. 36, no. 4, pp. 511–526, Sep. 2020, ISSN: 2446-4740. DOI: 10.1007/s42600-020-00089-1.

Grants

2026 – 2031 **Discovery Grant: 192,500$**
Title: *Locally Private Representation Learning for Responsible Artificial Intelligence*
Role: Principal Investigator

2025 – 2029 **Chair Program: 500,000€**
Title: *Responsible AI: Design, Regulation, and Conformity (RAIDAC+)*
Role: Co-Chair with Prof. Theodore Christakis

2025 – 2029 **ANR AAPG 2024 – JCJC (Young Researcher)[2]: 347,310€**
Title: *Aligning Privacy, Utility, and Fairness for Responsible AI (AI-PULSE)*
Role: Principal Investigator

2024 – 2026 **Inria Associated Team[3]: 32,000€**
Title: *Algorithmic Auditing of Privacy and Fairness (AUDIT-PAIR)*
Role: Principal Investigator (Inria side)

2023 – 2027 **ANR AAPG 2023 – PRCE (Academic and Industry Collaboration): 338,000€**
Title: *Making PostgreSQL Differentially Private for Transparent AI (DIFPRIPOS)*
Role: Member

2024 – 2025 **Open call to sustain the development and promotion of AI: 10,000€**
Title: *Exploring the Interplay of Differential Privacy and Fairness in ML*
Role: Principal Investigator

[2]Appel à Projets Générique de l'Agence Nationale de la Recherche (ANR).
[3]This program funds mobility between Inria and an international research group for 3 years.

Awards

2025 **Notable Reviewer Award** at USENIX Security 2025

2024 **Best Reviewer Award** at PETS 2024

2023 **Best Paper Award** at DBSec 2023 for the paper "(Local) differential privacy has no disparate impact on fairness"

2021 **Ph.D. Student Mobility Grant (2,300€)** from the University Bourgogne Franche-Comté (UBFC) to visit the Université du Québec à Montréal (UQAM)

2013 **UNEMAT Scholarship (19,200R$ – 400R$/month)** to be a collaborator in the Formation of Cooperative Cells (FOCCO) program for 4 years (2013 – 2016)

Teaching

Winter 2027 **Systems Security**
Undergraduate, ÉTS Montréal

Summer 2026 **Data Privacy**
Undergraduate, ÉTS Montréal

Spring 2021 **Privacy for IoT**
Graduate, Université Bourgogne Franche-Comté

Winter 2020 **Privacy for IoT**
Graduate, Université Bourgogne Franche-Comté

Students

PhD Students

- **Marwa El Kamil** (2025 - Present), INSA de Lyon & ÉTS Montréal
  Research Topic: *Privacy Risks in Foundation Models and AI Agents*
  Co-supervisors: Ulrich Aïvodji, Antoine Boutet
- **Zhan Xu** (2025 - Present), Université Grenoble Alpes (UGA) & ÉTS Montréal
  Research Topic: *Private and Fair Machine Learning*
  Co-supervisors: Ulrich Aïvodji, Claude Castelluccia
- **Imen Bajar** (2025 - Present), INSA de Lyon & ÉTS Montréal
  Research Topic: *Assessment and reduction of re-identification risk from internal brain tissues of brain MRIs*
  Co-supervisors: Mohamed Maouche, Carole Frindel, Antoine Boutet
- **Karima Makhlouf** (2022 - 2024), Institut Polytechnique de Paris (IPP)
  Thesis: *Advancing Ethical and Responsible AI: Exploring Fairness, Privacy, and Explainability through Causal Perspectives* [link]
  Co-supervisors: Catuscia Palamidessi

Master Students

- **Vinícius Gabriel Angelozzi** (2025), Université Grenoble Alpes (UGA)
  Research Topic: *Benchmarking Fairness-Aware Learning on Differentially Private Synthetic Tabular Data*
- **Annika Sauer** (2025), Karlsruhe Institute of Technology (KIT) & Université Grenoble Alpes (UGA)
  Thesis: *Measurement of Attack Resilience of Differential Privacy* [link]
  Co-supervisors: Patricia Guerra-Balboa
  UFA/DFH Prix d'Excellence

Services

Program Committee

2027 VLDB, PETS, FAccT
2026 CCS, AAAI
2025 CCS, PETS, USENIX Security, ICLR
2024 CCS, PETS, IJCAI, FAccT, ICLR
2023 NeurIPS, FAccT, ECML PKDD, CCS, ICLR
2022 ECML PKDD

Conference & Workshop Organization

2026 7th Summer school on deep learning for medical imaging (DLMI 2026), Canada
2024 14th Atelier sur la Protection de la Vie Privée (APVP 2024)[4], France
2023 13th Atelier sur la Protection de la Vie Privée (APVP 2023), France
2022 1st Comète Workshop on Ethical AI, France
2017 II Semana da Animação, Modelagem e Automação, Brazil
2014 I Semana da Faculdade de Ciencias Exatas, Brazil

Talks

2026 **Keynote Speaker:** *"Privacy Protection in Medical Imaging"* at 7th Summer School on Deep Learning for Medical Imaging (DLMI 2026)

2025 **Keynote Speaker:** *"Interplay of privacy and fairness in machine learning"* at 2nd Unite! Research School (URS 2025)

Keynote Speaker: *"Intersections of Fairness and Privacy: A Local Differential Privacy Perspective"* at 9th GDR RSD/ASF Winter School on Distributed Systems & Networks 2025

2024 **Invited Tutorial:** *"Securing Data with Local Differential Privacy: Concepts, Protocols, and Practical Applications"* at Selected Areas in Cryptography Summer School 2024

Invited Talk: *"Differentially Private Data Publishing: An Overview"* at Institut national d'études démographiques (Ined)

2023 **Invited Talk:** *"Locally differentially private protocols for frequency estimation of longitudinal data"* at Groupe de travail Protection de la Vie Privée (GT-PVP)

2022 **Invited Tutorial:** *"A Brief Introduction to Local Differential Privacy"* at SYSTOPIA Lab

Invited Talk: *"Data anonymization and Artificial Intelligence Models (in Portuguese)"* at Hospital Risoleta Tolentino Neves (HRTN)

2021 **Invited Talk:** *"Improving Utility and Privacy in Multidimensional Frequency Estimates Under Local Differential Privacy"* at Université du Québec à Montréal (LATECE Seminar)

Skills

Languages Strong reading, writing and speaking for English, French, Spanish, and Portuguese
Coding Python, Java, Matlab & Simulink, Visual Basic, LaTeX
Databases MySQL, PostgreSQL

[4]Translates to "Workshop on Privacy". This is the national French workshop from the privacy-preserving community GT-PVP. More info at https://www.univ-orleans.fr/lifo/evenements/GT-PVP/.

Skills (continued)

Misc.	Academic research, teaching, supervising, consultation, and publishing